REC: [short title, to be decided after planning]

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Abstract

[Abstract (ca. 100 words), to be written at the end.]

Keywords: Gamification, Game-based learning, Student response systems, Learning analytics, xAPI

Required Metadata

Current code version

1. Motivation and significance

Game-based student response systems such as Kahoot! have become common in higher education, where they are used to increase participation and to give students immediate feedback [1]. At the University of Cádiz, a teaching-innovation project running since the 2017/18 academic year in the course *Herramientas Multimedia y Animación*, now *Animación y Nuevas Tecnologías en Publicidad*, scores the quizzes held after each lecture by correctness and response speed, and the resulting rankings reward students in the final exam [2]. Encouraging participation is therefore no longer the main challenge. What teachers still lack is an analysis of what each student has actually learnt, built from their answers during the quiz and kept in the teacher's hands rather than the provider's.

The tool presented here grew out of this practice. In late 2019, the free plan of Kahoot! for higher education was limited to 50 players per live game, and in 2021 to 10 [3, 4], which made it unusable in the course *Fundamentos Técnicos en la Publicidad Digital*, with about 150 students. Moving the quizzes to Moodle, the institutional learning management system, solved the

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Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0 (`https://doi.org/10.5281/zenodo.23594648`)
C2	Permanent GitHub link to code/repository used for this code version	`https://github.com/lorenagmad/REC`
C3	Legal Code License	AGPL-3.0-or-later
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	JavaScript (ECMAScript 2020), HTML5, CSS3; Web Crypto, IndexedDB and Web Audio APIs; no third-party libraries
C6	Compilation requirements, operating environments & dependencies	A current web browser (Chrome, Edge, Firefox or Safari) on a computer, tablet or phone; any static web host with HTTPS, such as GitHub Pages, to publish contests
C7	If available Link to developer documentation/manual	`https://github.com/lorenagmad/REC/blob/main/docs/GUIDE.md`
C8	Support email for questions	lorena.gutierrez@uca.es

Table 1: Code metadata (mandatory)

problem of scale but not that of scoring, because Moodle quizzes do not rank students by correctness and response speed. Each quiz had to be downloaded, filtered and scored in a spreadsheet, and the ranking then published in the virtual campus by hand. This took 15 to 30 minutes per quiz, that is, 3 to 6.5 hours per semester for the 13 quizzes of the animation course.
Commercial game-based response systems such as Kahoot! [5], Wayground (formerly Quizizz) [6], Socrative [7], Blooket [8], Gimkit [9] and Mentimeter [10] let students join without an account, and several of them award points for speed. However, they store students' answers on the provider's servers, and their free plans limit the number of players or the export of results (Table 2). The Moodle quiz [11] keeps the data within the institution and imports Moodle XML, but it requires student accounts and does not score speed. Open-source alternatives such as ClassQuiz [12] and QuizDock [13] can be self-hosted, but they need a server with a database. Response times also carry information about learning: answers that reflect misconceptions tend to be given faster than correct ones [14], and in gamified quizzes the response time of wrong answers helps to tell guesses from weak recall [15]. According to their documentation, none of these tools combines speed and correctness in its analysis of students' answers.

Table 2: Comparison with related tools, according to their documentation (October 2026). Speed: points depend on response time; Static: runs as static files, with no application server or database; No cloud: answers are not stored on the provider's servers; Moodle XML: imports Moodle XML questions; Times: exports each student's response time per question; Speed×acc.: built-in analysis combining speed and correctness; Retention: built-in analysis across sessions; xAPI: exports xAPI statements; 150 free: a free game for about 150 students. ✓ yes; ○ partial; × no; ? not documented.

Tool	Speed	Static	No cloud	Moodle XML	Times	Speed×acc.	Retention	xAPI	150 free
REC	✓	✓	✓	✓	✓	✓	✓	✓	✓
Kahoot! [5]	✓	×	×	×	○[a]	×	○[a]	?	×
Wayground [6]	○[a]	×	×	?	○	×	?	?	×
Socrative [7]	×	×	×	×	×	×	?	?	×
Moodle quiz [11]	×	×	○[b]	✓	×	×	×	×	✓
Blooket [8]	✓	×	×	×	○[c]	×	?	?	×
Gimkit [9]	×	×	×	×	×	×	×	?	✓
Mentimeter [10]	✓	×	×	?	?	×	?	?	○

[a] Paid plans only. [b] Stored on the institution's server. [c] Average time per question only.

REC was developed under three requirements: (i) to reuse the questions teachers already have in Moodle, through the import of Moodle XML; (ii) to run in any browser without student accounts or servers, with contests published encrypted on a static website and opened with a password; and (iii) to give the teacher an analysis of each student's answers, combining speed and correctness, with the data held by the teacher. The teacher prepares the contests in the studio and publishes them encrypted. After the lecture, all students play at the same time, each on their own device (phone, tablet or laptop), with the password given in class; nothing is projected during the quiz. At the end, each student obtains a ticket with a verification code, with which the studio detects tickets that have been modified, and a results file in xAPI format [16], which they submit through the virtual campus. The teacher then verifies the tickets, builds and publishes the ranking, and analyses the results. The same tool supports graded continuous assessment, with rankings that reward students in the final exam, and ungraded diagnostic quizzes held before a topic is taught, which also record attendance.

2. Software description

[Planned: Architecture (two self-contained files: studio and studio floor), functionalities (including point-on-image questions, practice mode and extra time, as features rather than exclusive ones), and how AI assistance in development is reported. State the limit of the ticket check: without a server, a student who recomputes the code in the browser's developer tools is not detected.]

2.1. Software architecture
2.2. Software functionalities
3. Illustrative examples

[Planned: To be negotiated.]

4. Impact

[Planned: To be negotiated. Includes the REC → ranking → Aspa personalised-exam workflow [17] and the second-semester trial (results in June 2027).]

5. Conclusions

[Planned: To be negotiated. Future work includes bundling the typefaces with the tool instead of loading them from Google Fonts.]

Acknowledgements

[Optional.]

References

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Current executable software version

REC is a web application interpreted by the browser, so there is no compiled executable. The studio floor is available at `https://lorenagmad.github.io/REC/` and the studio at `https://lorenagmad.github.io/REC/studio.html`; the demonstration contest opens with the password `rec demo`.